Practical no.1: Implement DDL statement. Create table, Modify table, Drop table	Date:

```
CREATE TABLE STUDENT6( ROLL_NO INT, NAME VARCHAR(20),
MARKS INT);
SELECT *FROM STUDENT6;

INSERT INTO STUDENT6 VALUES(1, 'OM', 95);
INSERT INTO STUDENT6 VALUES(2, 'PRATHAMESH', 80);
INSERT INTO STUDENT6 VALUES(3, 'YASH', 85);
INSERT INTO STUDENT6 VALUES(4, 'KARTIK', 70);
SELECT *FROM STUDENT6;

ALTER TABLE STUDENT6 ADD RANKS INT;
SELECT *FROM STUDENT6;

ALTER TABLE STUDENT6 DROP COLUMN RANKS;
SELECT *FROM STUDENT6;
```

Practical no.2: Implement DML Statement. • Adding/Modify/Delete data using Insert/ Update/ Delete.	Date:

Insert Data:

```
CREATE TABLE STUDENT
(
    ROLL_NO INT,
    NAME VARCHAR(20),
    MARKS INT,
    EMAIL VARCHAR(40)
);

INSERT INTO STUDENT VALUES (1,'DEEP',85,'DEEP@GMAIL.COM');
INSERT INTO STUDENT VALUES (2,'OM',90,'OM@GMAIL.COM');
INSERT INTO STUDENT VALUES (3,'ROHAN',98,'RO@GMAIL.COM');
INSERT INTO STUDENT VALUES (4,'RAHUL',75,'RAHUL@GMAIL.COM');
INSERT INTO STUDENT VALUES (5,'SOHAM',88,'SOHAM@GMAIL.COM');

SELECT * FROM STUDENT;
```

Update Data:

```
UPDATE STUDENT SET NAME='SOHAM' WHERE ROLL_NO=5;
UPDATE STUDENT SET EMAIL='OM@GMAIL.COM' WHERE NAME='OM';

SELECT * FROM STUDENT;
```

Delete Data:

```
DELETE FROM STUDENT WHERE ROLL_NO=3;
DELETE FROM STUDENT WHERE ROLL_NO=4;

SELECT * FROM STUDENT;
```

Practical no.3: Implement following Constraints. • NULL and NOT NULL, Primary Key Constraint, Foreign Key Constraint • Unique Constraint, Check Constraint, Default Constraint.	**Date:**

```
create table department16(ID int primary key,
NAME varchar(50)not null,
HOD varchar(50)not null,
);

 insert into department16 values(11,'Harsh', 'Sweta Phegde');
insert into department16 values(12,'Pavan', 'Sweta Phegde');
insert into department16 values(13,'Aryan', 'Sweta Phegde');
insert into department16 values(14,'Ganesh', 'Sweta Phegde');
insert into department16 values(15,'Deep', 'Sweta Phegde');

select *from department16;

create table employee16(
ID int primary key,
NAME varchar(50)not null,
Email varchar(50) unique,
salary money check(salary>=6000),
city varchar(45) default'jalgaon',
department16id int foreign key references
department16(id)
);
select *from employee16;

select *from employee16;
insert into employee16 values(1,'Harsh', 'harsh07@gmail.com',
20000,'PUNE',11);
insert into employee16 values(2,'Pavan', 'pavan007@gmail.com',
50000,'Jalgaon',12);
insert into employee16 values(3,'Aryan', 'aryan44@gmail.com',
10000,'Nasik',13);
insert into employee16 values(4,'Ganesh', 'ganesh42@gmail.com',
16000,'Nagpur',14);
insert into employee16 values(1,'Harsh', 'harsh07@gmail.com',
20000,'Mumbai',15);
```

Practical No 4: Implement following clauses. • Simple select clause • Accessing specific data with Where Clause • Ordered By/ Distinct/Group By Clause.	Date:

```
CREATE TABLE EMPLOYEES(
EMPID INT PRIMARY KEY,
EMPNAME VARCHAR(255),
EMAIL VARCHAR(255),
CITY VARCHAR(255),
SALARY INT,
DEPARTMENTNAME VARCHAR(255)
);

SELECT * FROM EMPLOYEES;


INSERT INTO EMPLOYEES
VALUES(1,'DEEP','DEEP2@GMAIL.COM','NASHIK',50000,'MANAGEMENT');
INSERT INTO EMPLOYEES
VALUES(2,'MAYUR','MAYUR@GMAIL.COM','PUNE',40000,'CS');
INSERT INTO EMPLOYEES
VALUES(3,'RUPESH','RUPESH@GMAIL.COM','MUMBAI',20000,'PHYSICS');
INSERT INTO EMPLOYEES
VALUES(4,'JAY','JAY@GMAIL.COM','NAGPUR',30000,'CS');
INSERT INTO EMPLOYEES
VALUES(5,'SAI','SAI@GMAIL.COM','PUNE',60000,'MANAGEMENT');

SELECT * FROM EMPLOYEES;

SELECT EMPID,EMPNAME,CITY,SALARY,DEPARTMENTNAME FROM EMPLOYEES;


SELECT TOP 5 EMPID,EMPNAME,CITY,SALARY,DEPARTMENTNAME FROM EMPLOYEES
ORDER BY SALARY DESC;


SELECT * FROM EMPLOYEES WHERE SALARY<=30000;

SELECT * FROM EMPLOYEES WHERE SALARY>=15000 AND SALARY<=40000;

SELECT * FROM EMPLOYEES WHERE DEPARTMENTNAME='CS' OR
DEPARTMENTNAME='PHYSICS';

SELECT * FROM EMPLOYEES WHERE DEPARTMENTNAME IN
('CS','PHYSICAL','MANAGEMENT');


SELECT * FROM EMPLOYEES WHERE SALARY BETWEEN 20000 AND 50000;
```

Practical No 5: Aggregate Functions – AVG, COUNT, MAX, MIN, SUM, CUBE	Date:

```
CREATE TABLE EMPLOYEES12(
ID INT,
NAME VARCHAR(20),
AGE INT,
SALARY INT
);

SELECT *FROM EMPLOYEES12;

INSERT INTO EMPLOYEES12 VALUES(1001, 'MAHESH', 25, 25000);
INSERT INTO EMPLOYEES12 VALUES(1002, 'DEEP', 19, 20000);
INSERT INTO EMPLOYEES12 VALUES(1003, 'CHETAN', 21, 21000);
INSERT INTO EMPLOYEES12 VALUES(1004, 'DARSHAN', 20, 19000);
INSERT INTO EMPLOYEES12 VALUES(1005, 'KETAN', 30, 35000);


SELECT COUNT(*)FROM EMPLOYEES12;

SELECT MIN(AGE)FROM EMPLOYEES12;


SELECT MAX(AGE)FROM EMPLOYEES12;

SELECT AVG(AGE)FROM EMPLOYEES12;

SELECT SUM(SALARY)FROM EMPLOYEES12;
```

Name: Chetan Sanjay Dongare	Roll No: 29
Practical No 6: Implement all string functions.	Date:

```
select LEN('deep');

select UPPER('rohan');

select lower('MOHAN');

select REVERSE('Hey there');

select REPLICATE('sql', 4);

select ('sql') +space(2)+('server');

select SUBSTRING('sql server in  dbms', 15, 4);

select REPLACE('Hey there', ' Whats Up', '!');

select LEFT('sql server', 6);

select RIGHT('sql server', 7);

select LTRIM('        HELLO');

select RTRIM('HELLO        ');

select CHARINDEX('TO', 'Om go to school');

select PATINDEX('%USE%', 'It in use sql server');

select CONCAT('HELLO There', '   ', 'What are u doing');
```

Practical 7:- Implement Date and Time Functions	Date:

```
select current_timestamp;

select dateadd(month,2,'2026/02/5') as dateadd;

select dateadd (year,2,'2024/03/3') as dateadd;

select datediff (month,'2021/2/3','2022/3/4') as datediff;

select datediff (year,'2021/2/3','2022/3/4') as datediff

select datediff (hour,'2026/2/4 11:00','2026/2/5 12:30') as datediff;

select datename (yy,'2025/3/5') as datepartstring;

select day ('2026/2/7 08:00') as dateday

select month ('2025/3/4 06:00' ) as datemonth;

select getdate();

select getutcdate ();


select isdate('2024/5/6')as validdate;
```

Implement use of UNION, INTERSECTION, SET DIFFERENCE.	Date:

```
create table employee1(
id int primary key,
name varchar(20) not null
);

insert into employee1 values ( 1 ,'Mohit');
insert into employee1 values ( 2 ,'Pavan');
insert into employee1 values ( 3 ,'Bhavesh');
insert into employee1 values ( 4 ,'himanshu');

select * from employee1
create table employee2(
id int primary key,
name varchar(20) not null
);
insert into employee2 values ( 1 ,'aaditya');
insert into employee2 values ( 2 ,'Pavan');
insert into employee2 values ( 3 ,'Dhiraj ');
insert into employee2 values ( 4 ,'Rushi');

select * from employee2

select * from employee1
union
select * from employee2

select * from employee1
union all
select * from employee2

select * from employee1
intersect
select * from employee2

select * from employee1
except
select * from employee2

select * from employee2
except
```

```sql
select * from employee
```

Name : Chetan Sanjay Dongare	Roll.no : 29
Practical 10: Implement practical performing different operations on a view.	Date :

```sql
CREATE TABLE STUDENT36(
StudentID INT ,
NAME VARCHAR(50),
Course VARCHAR(50),
Marks INT
);
Select *from student36;


INSERT INTO STUDENT36 VALUES (1, 'Pavan', 'BCA', 85);
INSERT INTO STUDENT36 VALUES (2, 'Deep', 'BCA', 80);
INSERT INTO STUDENT36 VALUES (3, 'Rahul', 'BCA', 60);
INSERT INTO STUDENT36 VALUES (4, 'Chetan', 'BCA', 50);
INSERT INTO STUDENT36 VALUES (5, 'Om', 'BCA', 95);


CREATE VIEW BCA_Student36 AS
Select studentID, Name, Course, Marks
From student36
where course = 'BCA';

Select * from BCA_student36;

Insert into BCA_student36 VALUES(6, 'Ram', 'BCA',92);

UPDATE STUDENT36 SET Marks=55
Where Name='Chetan';

Delete from BCA_student36
Where StudentID =5;

Drop View BCA_Student36
```

Practical1: Implement use of Procedure.	Date :

```sql
CREATE TABLE Student_bca1
(
   StudentID INT PRIMARY KEY,
   Name VARCHAR(50),
   Age INT,
   Course VARCHAR(50)
);

SELECT * FROM Student_bca1;

INSERT INTO Student_bca1 VALUES(1, 'Om', 'Computer Application');
INSERT INTO Student_bca1 VALUES(2, 'Deep', 'Computer Science');
INSERT INTO Student_bca1 VALUES(3, 'Chetan', 'Engineering');

CREATE PROCEDURE AddStudent_bca1
   @StudentID INT,
   @Name VARCHAR(50),
   @Age INT,
   @Course VARCHAR(50)
AS
BEGIN
   INSERT INTO Student_bca(StudentID, Name, Age, Course)
   VALUES(@StudentID, @Name, @Age, @Course);
END;

EXEC AddStudent_bca1 4, 'Sneha', 23, 'Mechanical';
EXEC AddStudent_bca1 5, 'neha', 24, 'Computer';

CREATE PROCEDURE ShowStudent_bca1
AS
BEGIN
   SELECT * FROM Student_bca1;
END;

EXEC ShowStudent_bca1;

CREATE PROCEDURE UpdateCourse
   @StudentID INT,
   @Course VARCHAR(50)
AS
BEGIN
   UPDATE Student_bca1
   SET Course = @Course
```

```sql
    WHERE StudentID = @StudentID;
END;

EXEC UpdateCourse 2, 'Data Science';
EXEC ShowStudent_bca1;


CREATE PROCEDURE DeleteStudent
    @StudentID INT
AS
BEGIN
    DELETE FROM Student_bca1
    WHERE StudentID = @StudentID;
END;

CREATE PROCEDURE DeleteName
    @Name VARCHAR(50)
AS
BEGIN
    DELETE FROM Student_bca1
    WHERE Name = @Name;
END;

EXEC DeleteName 'neha';
EXEC ShowStudent_bca1;
```